Advancing Coral Reef Bathymetry: A GAN-Augmented and Stratified CNN Analysis of Fused ICESat-2 and Sentinel-2 Dataset

Ziyao Chen, Li Wang, Wei Feng, Yan Gu, Jin Li, and Ya Ping Wang

***Abstract*—High-resolution bathymetry mapping of coral reefs is essential to morphodynamic study of reef habitats, assisting reef monitoring and conservation under global climate change. However, the accuracy of conventional satellite-derived bathymetry (SDB) is reduced at depths over 15 m with optical signal attenuation and training data insufficiency. To address this gap, here, we present an approach that synergizes ICESat-2 advanced topographic laser (ATL24) photon-counting LiDAR data with Sentinel-2 multispectral imagery. A generative adversarial network (GAN) is implemented to offset dataset deficiency at deeper depths, and a stratified convolutional neural network (CNN) is adapted to distinct optical-depth regimes. Bathymetry derived at Jiuzhang Atoll is in good agreement with the in situ multibeam measurements, with a mean absolute error (MAE) of 0.75 m and a root-mean-squared error (RMSE) of 10% of the present maximum depth of 19 m, validating the effectiveness of GAN-driven sample synthesis to make up measurement inadequacy, and the enhancement of model generalizability across a wide depth range by stratified CNN. This approach could be applied to bathymetry mapping of coral reefs worldwide at depths of 15–30 m, where biodiversity generally increases the most with multisource satellite observations.**

***Index Terms*—Convolutional neural network (CNN), coral reef bathymetry mapping, data augmentation, generative adversarial network (GAN), ICESat-2, satellite-derived bathymetry (SDB), Sentinel-2.**

I. Introduction

DESPITE covering less than 1% of the ocean floor, coral reefs are renowned for their remarkable biodiversity and productivity, providing sustenance for over a quarter of marine species and more than 450 million people [1], [2]. Most coral reefs exist in waters less than 50-m depth, with a low biodiversity near the sea surface and an increasing biodiversity to a maximum between 15- and 30-m depth [2]. High-resolution bathymetry mapping of coral reefs is the foundation of hydrodynamic and geomorphic study of reef habitats [3], [4], facilitating ecological monitoring and conservation under global climate change [5], [6], as well as supporting navigational safety [7], [8]. While methods of shipborne multibeam sonar and airborne bathymetric LiDAR deliver high-precision bathymetric information, their application to large-scale and/or regular mapping of submarine topography is restricted in coral reef areas owing to the high operation costs, logistical complexity, and limited spatial–temporal coverage [9], [10], [11], [12].

Alternatively, satellite-derived bathymetry (SDB) methods are used to map the bathymetry of shallow waters without physically being on-site, leveraging the global coverage and frequent revisit times of multispectral satellites such as Sentinel-2 [13], [14]. The launch of ICESat-2, which is equipped with the advanced topographic laser (ATL) altimeter system, advances the global application of SDB calibration. High-accuracy along-track depth measurements are provided by the photon-counting LiDAR, serving as a global reference to sites where in situ bathymetric measurement is unavailable [15]. The promoted SDB accuracy with the combination of ICESat-2 data and optical imagery has been demonstrated in many recent studies [16], [17], [18], [19]. The latest release of the standardized ATL24 product supports a worldwide consistent bathymetric reference, enhancing the interoperability and reproducibility of SDB workflows [20].

Despite these advances, there is a pronounced decline in the accuracy of data-driven SDB with increasing water depth, primarily due to the rapid attenuation of light intensity and thus the optical signal availability of ICESat-2, and the deficiency of reference data in deeper shallow-water zones over 15 m [17], [20]. Data-driven SDB with deep learning approaches yield overfitting results with high uncertainty [14]. Such a limitation of validation data deficiency is particularly obvious in the bathymetry mapping of coral reef areas, where field surveys are sparse [19]. It is thus crucial to enhance reliable training data in deeper zones of shallow waters and to adapt the modeling framework to variable optical environments.

Generative adversarial network (GAN), which generates a physically accountable dataset by learning the primary data distributions, could be a potential solution to the limitation of data scarcity [21]. It is found that GAN works well in image enhancement in remote sensing [22], [23]. Yet, few

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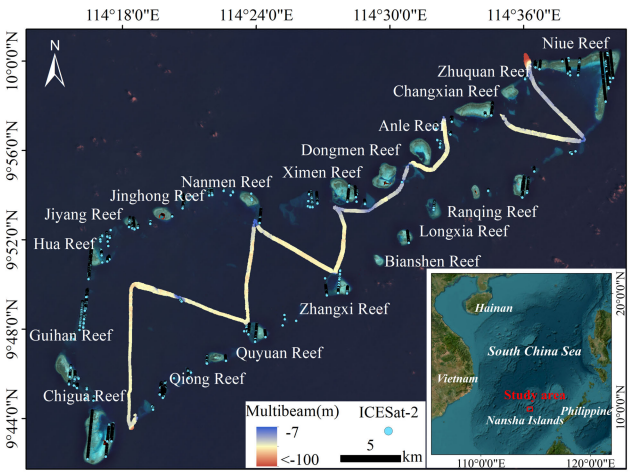


Fig. 1. Map of the exemplified study area of Jiuzhang Atoll in South China Sea. The basemap is a Sentinel-2 image acquired on March 15, 2019 with flare correction. The route of the in situ multibeam bathymetric survey in 2016 that is used for model evaluation is marked by the path with color gradient representing water depths in meters. The blue points denote the resampled ICESat-2 measurements.

exploration has been conducted on the application of GAN to feature-level augmentation in SDB, refining depth-specific spectral relationships.

In this study, the compounded limitation of data deficiency and depth-varying optical attenuation for SDB reliability within deeper shallow-water zones (>15 m) is addressed. GAN for feature-level augmentation of SDB guided by physical plausibility constraints is committed. The results are coupled with depth-stratified convolutional neural networks (CNNs) to generate a robust bathymetry mapping of the coral reef. In Section II, Jiuzhang Atoll, as an exemplified study area, is briefly introduced. In Section III, the methodology is described in detail. First, the performance of different inversion models of satellite-based datasets is evaluated across various depth ranges in Jiuzhang Atoll. Second, a GAN-based data augmentation strategy is applied to the processed satellite-based depth data, whose impact on model accuracy and stability is assessed. Third, a stratified CNN model is implemented for three depth intervals with distinct optical intensity to improve bathymetric retrieval in a further step. In Sections IV and V, the bathymetric results are presented and compared with multibeam survey data and a public benchmark dataset. The performance and practical applicability of this approach are evaluated and discussed.

II. Study Area

Jiuzhang Atoll (114°14′E–114°41′E, 9°42′N –10°02′N) is located in the northern part of the Nansha Islands, which is the largest coral reef group in South China Sea (Fig. 1). Similar to most atolls in the Nansha Islands, Jiuzhang Atoll is characterized by a well-defined spindle-shaped rim surrounding the central lagoon of tens of meters deep, with a major axis extending in a generally northeast–southwest direction. Such an orientation pattern is attributed hypothetically to either tectonic-controlled seafloor geometry or East-Asian monsoon activity [24], [25]. The margin of Jiuzhang Atoll is comprised of both shallow parts (reef crest and flat) that are subaerial at low tide and submerged banks at depths ranging from several meters to even tens of meters below sea level. The widespread spur-and-groove systems across the submerged reef zone generate a distinct and complex topographic gradient [26]. Coral reefs are characterized by clear waters with low suspended sediment concentrations, which are conducive to SDB. Yet, accurate seafloor detection in deeper zones over 15-m depth, where abundant reef biodiversity generally occurs, could be impeded by light intensity attenuation, where reference data from ICESat-2 are consequently sparse, as mentioned above. This collectively makes the atoll an ideal object for testing the bathymetric inversion method that combines GAN augmentation and CNN stratification proposed in this study.

III. Methodology

A. Observation Data and Preprocessing

Sentinel-2 Multispectral Instrument Level-2A surface reflectance products are acquired from the Copernicus/S2_SR_HARMONIZED collection via Google Earth Engine. A cloud-free scene on March 15, 2019 is selected in accordance with the in situ surveys. Four spectral bands—blue (B2, with a wavelength of 490 nm), green (B3, 560 nm), red (B4, 665 nm), and near-infrared (NIR, B8, 842 nm)—are retained at their native 10-m resolution to leverage their differential water penetration capabilities [27], [28]. To mitigate radiometric contamination from sun glint, a deep-water correction is applied using the GEBCO's global bathymetric datasets [29]. Pixels deeper than 100 m are used to establish a linear relationship between each visible-light spectral band and the NIR band. Regression coefficients are applied to remove glint effects across the scene.

A combined spectral–bathymetric masking approach is implemented to define optically shallow waters. The normalized difference water index (NDWI) using bands B3 and B8 initially identified water pixels [19]. This mask is further refined using the spatial distribution of ICESat-2 ATL24 seafloor points so that analysis is restricted to depths where bottom reflectance is detectable.

Dataset in 2018–2020 of ICESat-2 ATL24 Version 1 is obtained from the National Snow and Ice Data Center https://nsidc.org/data/atl24/versions/1 (accessed on August 8, 2025) as high-accuracy reference depths (Table A.1, Supplementary Material). Seafloor photons (class_ph = 40, Table A.1, see Supplementary Material) with confidence scores >0.8 and refraction-corrected orthometric heights <0 are retained, yielding 54 134 qualified depth estimates.

To align with Sentinel-2 resolution, ATL24 photons are aggregated within 10×10 m grid cells, with mean depth computed per pixel. After cloud screening and manual curation, 7404 spatially coregistered depth points are retained (Fig. 2), with 80% allocated for training and 20% for validation.

A multibeam echosounder survey across the Jiuzhang Atoll in November 2016 provided high-accuracy validation data, which are processed and referenced to the EGM2008 geoid to maintain vertical consistency with ATL24. To assess the potential impact of temporal mismatch between the multibeam survey (2016) and satellite observations (2018–2020),

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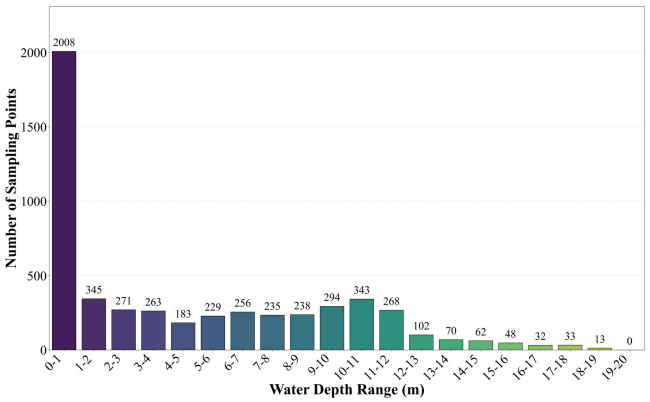


Fig. 2. Distribution of resampled ICESat-2 ATL24 depths.

historical tropical cyclone records from the IBTrACS database are examined [30]. No typhoon cyclones with maximum wind speed over 64 knots were recorded to pass within 100 km of the study area during this period (2016–2020). In addition, the HHU24SWDSCS dataset [19] and GEBCO_2025 grid [29] are incorporated for cross-comparison and deep-water masking, respectively.

B. Comparison of Bathymetric Inversion Models

Four bathymetric inversion models are implemented, and results are compared, including the Lyzenga model [31], random forest (RF) [32], backpropagation neural network (BPNN) [33], and CNN [34].

A linear multispectral model is introduced by Lyzenga to derive shallow-water bathymetry from optical remote sensing data, establishing a linear relationship between remote sensing reflectance and water depth [31]. The model coefficients are determined by minimizing the difference between estimated and measured depths. For Sentinel-2 imagery, the model is expressed as

$$\hat{H} = h_0 + \sum_{j=1}^{N} h_j R_{rsj} \tag{1}$$

where $\hat{H}$ denotes the predicted water depth; R_{rsj} is the above surface remote sensing reflectance of the jth band after atmospheric and geometric corrections; and h_0 and h_j are the intercept and regression coefficient of the linear model, respectively.

RF is a machine learning method composed of multiple decision trees. It utilizes multiple decision trees during training and prediction, with the final output determined by averaging the values produced by each decision tree [35], [36]. BPNN is a multilayer feedforward network that adjusts weights through error backpropagation to minimize prediction errors [37]. Here, we construct a BPNN with three input nodes and seven hidden neurons, employing a hyperbolic tangent activation function in the hidden layer. Model training utilizes the Adam optimizer with an initial learning rate of 0.05, a maximum of 1500 epochs, and a convergence tolerance of 1×10^{-5}. To prevent overfitting, an early stopping strategy is implemented, terminating training if the validation loss does not improve for 20 consecutive epochs.

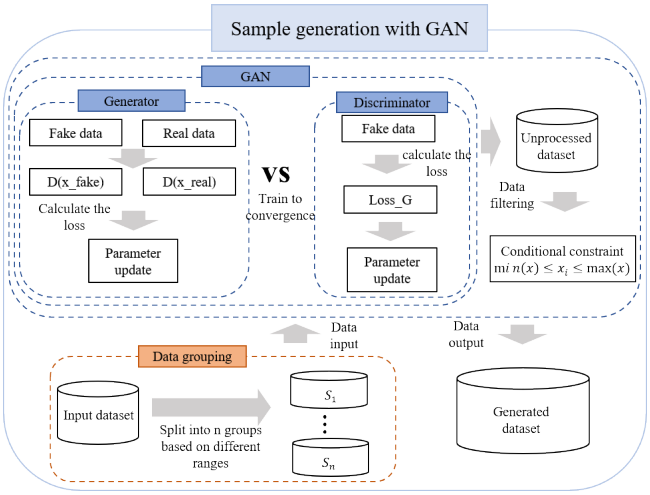


Fig. 3. Workflow of grouped GAN-based data generation.

A 1-D CNN is implemented for optical remote sensing bathymetry [38]. The model input consisted of the blue, green, and red reflectance bands (B2–B4) from preprocessed Sentinel-2 imagery. The network architecture comprises a convolutional layer with 64 filters (kernel size = 2), followed by a max-pooling layer and two fully connected layers with 50 neurons. The network contains 3493 trainable parameters and is optimized using the Adam algorithm. To enhance model robustness, a dropout layer with rate of 0.2 is applied during training.

C. GAN-Based Data Augmentation for Depth-Specific Sample Generation

To address critical sample scarcity over a 15-m depth range (Fig. 2), a GAN is employed with a grouped training strategy and output constraints for data augmentation. Samples within each group exhibit similar depth characteristics. The GAN framework consists of a generator (G) that learns collectively the distribution of spectral features and depths, and a discriminator (D) that distinguishes actual observations from synthetic samples (Fig. 3). The detailed GAN configuration is summarized (Table A.2, Supplementary Material). To improve training stability and avoid mode collapse, label smoothing and dropout regularization are applied, and training is conducted separately for different depth intervals. In addition, training is conducted on two depth groups of 15–17 m and 17–19 m, to avoid model bias or training instability owing to uneven sample distribution. Generated samples are filtered by physical plausibility constraints [see (2)] so that the synthetic results are physically meaningful and statistically consistent in terms of spectrum and depth

$$\min(x) \le x_i \le \max(x) \tag{2}$$

where x_i and x represent the generated feature-depth pairs and original data, respectively. This produced 800 synthetic samples, effectively balancing the training set (Fig. 3).

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D. Linear Interpolation-Based Data Augmentation

To isolate the contribution of GAN-based augmentation, we implement a simple linear interpolation baseline that generates synthetic depth–spectral samples x_{new} by convex combinations of two real samples within the same depth group

$$x_{\text{new}} = w \cdot x_i + (1 - w) \cdot x_j \tag{3}$$

where x_i and x_j denote two neighboring depth–spectral feature samples. w is a weighting factor uniformly sampled from [0, 1].

E. Stratified CNN Modeling for Various Depth Inversions

Sample distribution analysis reveals various data availability across depths. In the case of Jiuzhang Atoll, depths are divided into three intervals of 0–12 m with an ample data supply, 12–15 m with a moderate data supply, and 15–19 m a the least data supply. A unified model would be biased toward shallower regions, limiting accuracy at deeper waters. Stratification of CNN modeling is thus implemented. CNN training is conducted for shallow-depth zone of 0–12 m, mid-depth 0–15 m, and deep-depth 0–19 m. GAN-augmented samples are incorporated in the deep-depth zone 0–19 m. Each zone is inverted with the optimal modeling results, in order to optimize adaptation to depth-specific optical features.

F. Accuracy Assessment

Model performance is evaluated using the coefficient of determination R^2, root-mean-squared error (RMSE), and mean absolute error (MAE) across three depth intervals of 0–12 m, 0–15 m, and 0–19 m, respectively,

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - y)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2} \tag{4}$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - y)^2} \tag{5}$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^{n} |y_i - y| \tag{6}$$

where y_i is the observed depth, y is the predicted depth, $\bar{y}$ is the mean observed depth, and n is the sample count.

IV. Results

A. Comparison of Inversion Accuracy Across Models

To evaluate the performance and robustness of different bathymetric inversion approaches, we compare bathymetric results of the four models—the Lyzenga model, RF, BPNN, and CNN—based on ICESat-2 ATL24 depths and Sentinel-2 multispectral data. Given that most reference samples are distributed within 0–19 m, the depth of 19 m is set as the maximum retrieval limit.

In the 0–12-m range, all models achieved high accuracy (Fig. 4; Table I). Both RF and CNN attained an R^2 of 0.97 and an RMSE of 0.71 m, with MAE values of 0.42 and 0.45 m, respectively, outperforming the Lyzenga model with an RMSE of 0.93 m and an MAE of 0.60 m, and BPNN with an RMSE of 0.74 m and an MAE of 0.49 m. When the depth threshold is extended to 0–15 m, CNN exhibited a superior performance, achieving an R^2 of 0.96, an RMSE of 0.89 m, and an MAE of 0.53 m, in comparison with RMSE of 1.02 m for Lyzenga and RF models, and RMSE of 0.93 m for BPNN.

Across the full 0–19-m range, accuracy declines with all the four models due to the lack of reliable samples at depth beyond 15 m. Nonetheless, depths reproduced by CNN exhibit the smallest errors, with an R^2 of 0.94, an RMSE of 1.17 m, and an MAE of 0.62 m. The BPNN and RF models show comparable performance with an RMSE of 1.20 and 1.25 m, respectively, while the Lyzenga model exhibits a larger error with an RMSE of 1.28 m.

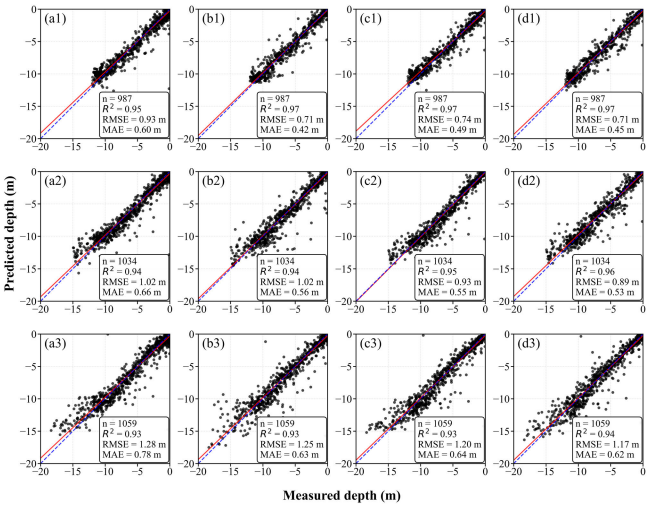


Fig. 4. Scatterplot comparison of multiple models across depth ranges. Bathymetric inversion performance of four models—Lyzenga, RF, BPNN, and CNN—on an independent ICESat-2 test set is illustrated. (a1)–(a3), (b1)–(b3), (c1)–(c3), and (d1)–(d3) Results of Lyzenga, RF, BPNN, and CNN, respectively. Subscripts 1–3 indicate validation on data subsets with maximum depths of 12, 15, and 19 m, respectively. The blue dashed line is the 1:1 reference line, and the red solid line is the linear regression fit to the data points.

TABLE I
QUANTITATIVE PERFORMANCE EVALUATION OF FOUR BATHYMETRIC INVERSION MODELS ON ICESAT-2 TEST SETS WITH DIFFERENT DEPTH RANGES

Depth range (m)	Model	R^2	RMSE (m)	MAE (m)
0-12	Lyzenga	0.95	0.93	0.60
	RF	**0.97**	**0.71**	**0.42**
	BPNN	0.97	0.74	0.49
	CNN	**0.97**	**0.71**	0.45
0-15	Lyzenga	0.94	1.02	0.66
	RF	0.94	1.02	0.56
	BPNN	0.95	0.93	0.55
	CNN	**0.96**	**0.89**	**0.53**
0-19	Lyzenga	0.93	1.28	0.78
	RF	0.93	1.25	0.63
	BPNN	0.93	1.20	0.64
	CNN	**0.94**	**1.17**	**0.62**

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TABLE II
DISTRIBUTION OF GAN-GENERATED SAMPLES

Depth range (m)	Group	Training samples	Generated samples
15-16	1	48	262
16-17	1	32	138
17-18	2	33	268
18-19	2	13	132

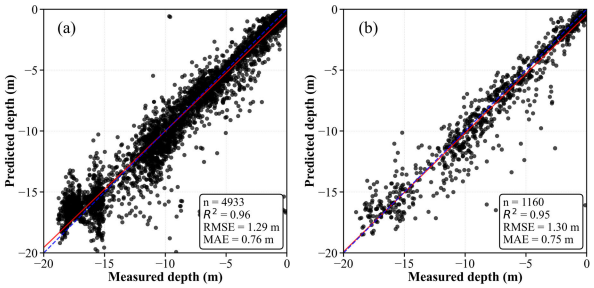


Fig. 5. Scatter plots of CNN-predicted depths in comparison with ICESat-2 observations after incorporating GAN-generated samples. (a) Training set and (b) independent validation set for depth of 15–19 m.

B. Results of GAN-Based Data Augmentation

The GAN model is trained on depth-partitioned sample groups to enforce structure-aware learning, with a range-control mechanism [see (2)] that ensures spectral and bathymetric consistency of the generated samples. Using this framework, 800 synthetic samples are generated to boost the 15–19-m depth range where direct observations are inadequate, as shown in Table II.

These 800 GAN-generated samples are incorporated into the stratified CNN training set to enhance data representation in depth over 15 m. The original ICESat-2 samples from the 15- to 19-m range serve as an independent test set to evaluate the effect of GAN augmentation. The bathymetry results trained with the GAN-augmented data present an improved accuracy across the depth range 0–19 m, with an R^2 of 0.95, an RMSE of 1.30 m, and an MAE of 0.75 m (Fig. 5).

To further examine how GAN-based augmentation improves model performance of bathymetry mapping, a feature space comparison is conducted for the deep-water interval (15–19 m). Fig. 6 shows the distribution of spectral features (B2–B4 and B8) for the original ICESat-2/Sentinel-2 samples and the GAN-synthetic samples. The generated datasets demonstrate a statistical distribution that is closely approximating to the original observations at depth range of 15–19 m while effectively increasing the sample density in sparsely populated regions of the feature space. So as to boost a better depth-spectral relationship, particularly at deeper depth.

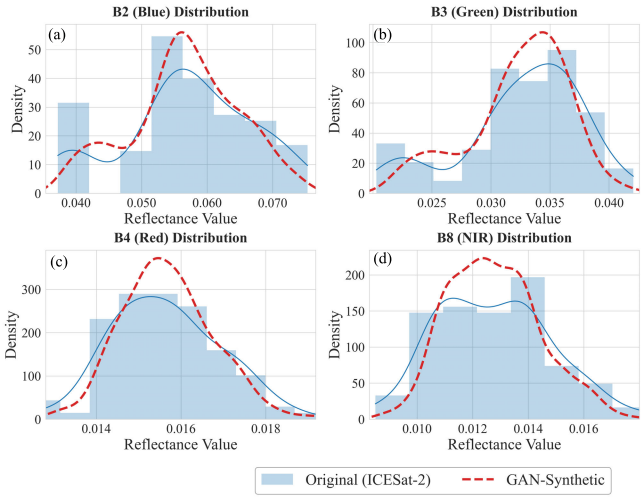


Fig. 6. Comparison of spectral feature distributions between original ICESat-2/Sentinel-2 observations and GAN-synthetic data across four bands. (a) B2 (blue), (b) B3 (green), (c) B4 (red), and (d) B8 (NIR). The blue histograms and solid lines represent the density distribution of the original samples, while the red dashed lines indicate the kernel density estimation of the GAN-synthetic samples.

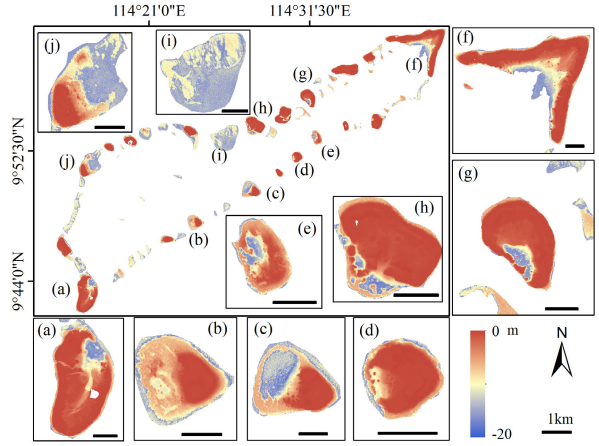


Fig. 7. Stratified bathymetry inversion results of Jiuzhang Atoll. (a)–(j) Enlarged views of selected regions within the atoll.

C. Bathymetry Inversion Results Based on Stratified CNN

To address the uneven distribution of ICESat-2 samples and the consequent decline in inversion accuracy, particularly in deeper regions such as 15–19 m in this case study, a stratified modeling approach is implemented. Building on the GAN-augmented and stratified CNN framework, a shallow model (0–12 m), a mid-depth model (0–15 m), and a deep model (0–19 m) incorporating synthetic samples for the 15–19-m range are developed, as described in Section III-D. The consistency analysis confirm satisfactory agreement between the stratified models within the 40-m buffer zones. The RMSE between adjacent predictions is 0.80 m at the shallow boundary (12 m) and 1.40 m at the deep boundary (15 m). The applied averaging strategy further ensured a seamless transition across depth intervals. The resultant bathymetric mapping of Jiuzhang Atoll is presented in Fig. 7.

To assess the bathymetric mapping dervied from the GAN-augmented and stratified CNN, a primary interpolation

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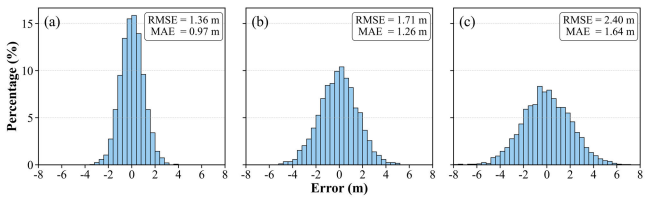


Fig. 8. Error distributions of the stratified CNN model validated against multibeam data. (a) 0–12 m, (b) 0–15 m, and (c) 0–19 m.

TABLE III
STRATIFIED VALIDATION WITH MULTIBEAM DATA ACROSS DIFFERENT MODELS

Depth range (m)	Model	RMSE (m)	MAE (m)
0-12	HHU24SWDSCS	1.87	1.57
	CNN(without data augmentation)	1.93	1.51
	CNN(with augmented samples)	1.36	0.97
0-15	HHU24SWDSCS	2.67	2.22
	CNN(without data augmentation)	1.92	1.40
	CNN(with augmented samples)	1.71	1.26
0-19	HHU24SWDSCS	-	-
	CNN(without data augmentation)	3.70	3.00
	CNN(with augmented samples)	2.40	1.64
15-19	HHU24SWDSCS	-	-
	CNN(without data augmentation)	3.82	3.44
	CNN(with augmented samples)	2.61	2.17

experiment is conducted using ordinary kriging to interpolate ICESat-2 bathymetric points. The results are compared with the independent multibeam measurements in 2016. Results show that spatial interpolation alone yields a poor performance, with an RMSE of 12.09 m and an MAE of 7.69 m. This is substantially worse than all machine learning models evaluated in this study, indicating that spatial interpolation of sparse along-track LiDAR points fails to reconstruct complex reef bathymetry.

Multibeam bathymetry observation in 2016 is used as an independent validation dataset to evaluate the approach. The spatial distribution of inversion errors across the three depth intervals is shown in Fig. 8. Errors are minimal in shallow regions and remain within acceptable bounds across the study area, despite a gradual increase with depth. Quantitative accuracy assessments are summarized in Table III. The stratified CNN with GAN augmentation consistently demonstrates the highest accuracy across all depth ranges. The most notable improvement is observed in the 15–19-m range, where RMSE and MAE are reduced to 2.40 and 1.64 m, from 3.70 to 3.00 m in the calculation with nonstratified CNN, respectively. The RMSE and MAE over the 0–19-m depth range of HHU24SWDSCS dataset, which has a maximum depth of about 15 m within this study area, are not evaluated. In the 15–19-m depth interval, the models exhibited significant error, with Lyzenga, RF, and BPNN yielding RMSEs of 4.31, 4.20, and 3.94 m, respectively. The standard CNN (without augmentation) performed similarly with an RMSE of 3.82 m. In contrast, the proposed GAN-augmented and stratified CNN achieved a significantly lower RMSE of 2.61 m. This represents a substantial accuracy improvement, reducing the error by 31.7% compared to the standard CNN and by 33.8%–39.4% compared to traditional machine learning models.

TABLE IV
ACCURACY OF BATHYMETRIC INVERSION UNDER DIFFERENT SEABED SLOPE CATEGORIES

Slope Range	RMSE (m)	MAE (m)
< 5°	1.49	1.23
5° – 15°	2.19	1.81
> 15°	2.78	2.45

To further examine the specific contribution of GAN-based feature learning, an ablation experiment is conducted by replacing GAN-generated samples with linearly interpolated spectral–depth features. Using this baseline strategy, the model achieved an RMSE of 3.10 m and an MAE of 2.42 m in the 15–19-m depth range. While this approach improves upon direct spatial interpolation, it remains inferior to the GAN-augmented model.

It is noted that validation against multibeam data yielded slightly higher RMSE and MAE values than those derived from ICESat-2 samples. First, ICESat-2 data exhibit a more uniform distribution across various depth ranges, whereas the multibeam validation points are limited to depths greater than 7 m. As water depth increases, the RMSE and MAE of the inversion model tend to rise due to enhanced light attenuation and reduced signal strength from the seafloor. The multibeam data were acquired in 2016, preceding the ICESat-2 measurements and Sentinel-2 imagery in 2018–2020, potentially introducing temporal mismatches. The overall accuracy of the results could be satisfactory.

D. Impact of Seabed Slope on Inversion Accuracy

To evaluate the model's adaptability to complex geomorphologies, the validation dataset is stratified into three terrain categories based on the true seabed slope derived from multibeam data: as flat (<5°), moderate (5°–15°), and steep (>15°). As shown in Table IV, the inversion accuracy varies across different with terrain complexity.

The model achieved high accuracy in flat terrains, which constitute the majority of the reef flat and lagoon areas, with an RMSE of 1.49 m and an MAE of 1.23 m. In contrast, the error is the largest at steep margins, with an RMSE of 2.78 m.

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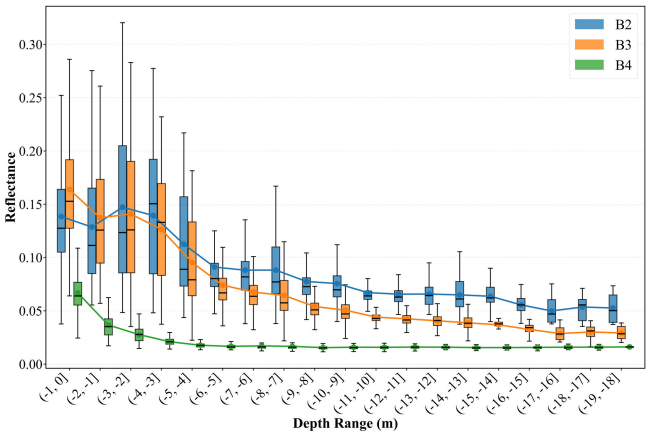


Fig. 9. Relationship between band reflectance and water depth.

V. Discussion

A. Influence of Water Depth on Model Inversion Capability

Water depth is a primary determinant of bathymetry accuracy with optical remote sensing bathymetry. With increasing depth, intensified absorption and scattering within the water column lead to exponential attenuation of bottom-reflected radiance received by the sensor [39], [40]. This signal decay complicates the relationship between spectral reflectance and depth, thereby reducing model fitting accuracy and predictive stability [41].

Although the CNN model achieved high accuracy in shallow waters (0–12 m) in this study, RMSE increases to approximately 3.0 m within the 15–20-m interval, indicating a clear decline in performance with depth (Table III). This trend, corroborated by multibeam validation, primarily stems from optical attenuation and the diminishing contribution of bottom reflectance to the overall signal. As illustrated in Fig. 9, the spectral reflectance in all bands exhibits an exponential decay as depth increases. Beyond 15-m depth, the penetration capacity of Sentinel-2's blue and green bands (B2 and B3) becomes limited, and the bottom signal constitutes only a minor fraction of the total reflectance, thereby weakening the spectral–depth relationship [14]. This physical limitation explains the observed instability of CNN predictions in deeper regions.

Furthermore, the uneven distribution of training samples exacerbates this issue. While abundant samples in depths <12 m enable the model to effectively learn nonlinear spectral–depth relationships, the scarcity of data beyond 15 m limits CNN performance. The performance of data-driven approaches hinges on sufficient and balanced training data to ensure reliability, consistent with prior SDB studies [42], [43].

It is demonstrated that the stratified modeling approach for separate CNN training and inversion in this study substantially improves model stability and accuracy in deeper waters (Table III). The shallow-water model effectively captured spectral variations under strong optical penetration, while the deep-water model is independently optimized for attenuation-dominated conditions. Overall, stratified modeling reduced the total RMSE by approximately 35% compared to the nonstratified baseline. This aligns with the findings that depth-stratified modeling effectively mitigates errors in deeper regions and enhances overall performance in nearshore environments [44].

Both the physical process of optical attenuation and the statistical issue of sample imbalance underscore the necessity of depth-stratified modeling for shallow-water bathymetric inversion. By integrating ICESat-2 LiDAR depths with Sentinel-2 multispectral imagery within a stratified framework, more accurate and reliable bathymetric mapping is obtained in optically complex coral reef areas at depths over 15 m.

B. Comparative Accuracy of Multiple Inversion Models

To assess the applicability and performance of different bathymetric inversion approaches, this study systematically compares four models—the Lyzenga model, RF, BPNN, and CNN—across multiple depth intervals (Section IV-A and Table I). Results indicate a general trend of improved accuracy with increasing model complexity and nonlinear representational capacity, with CNN consistently achieving the superior performance.

The Lyzenga model, as a log-linear empirical method, assumes a quasi-linear relationship between water-leaving reflectance and depth. It performs reliably in clear, shallow waters with homogeneous substrates, but its accuracy declines significantly in deeper areas or under complex optical conditions due to its inability to capture nonlinear attenuation effects [45]. In this study, the Lyzenga model yields acceptable RMSE in depths under 12 m, but errors increase markedly beyond this range.

In comparison, the BPNN model learns complex spectral–depth relationships through multilayer nonlinear mapping, offering stronger fitting capability than RF. Under favorable optical conditions and with adequate training samples, it effectively reconstructs radiative transfer processes. However, it remains highly sensitive to sample size and distribution, and is prone to overfitting or gradient instability—particularly in deeper zones with sparse samples, leading to increased prediction variance [46], [47].

The CNN model further advances this progression by incorporating spatial convolutional mechanisms that automatically extract high-order spectral–spatial features from multispectral input [48]. In our experiments, the CNN achieve the highest accuracy across all depth intervals, with the least RMSE and MAE among all the models. This performance advantage likely stems from the CNN's capacity to jointly learn local spectral patterns and contextual features, enabling a more robust approximation of complex underwater light fields. Nevertheless, its accuracy in the 15–19-m range remains constrained by inherent physical limitations—optical signal attenuation and the scarcity of reference data.

Overall, the progression from the Lyzenga model to the CNN reflects a technological shift from linear empirical formulations to spatially aware deep learning architectures. Machine learning and deep learning methods, with their enhanced nonlinear fitting and hierarchical feature extraction capabilities, more effectively model complex in water optical processes [49], demonstrating an effective approach to derive high-resolution, large-scale bathymetric inversion.

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C. Improvements to Bathymetric Inversion via GAN-Based Data Augmentation

In shallow-water bathymetric inversion, the quantity and distribution of training samples are critical to model performance and generalization capability [50]. To overcome this limitation, we introduce a GAN-based data augmentation strategy to expand the representation of deeper zones in the training set.

Experimental results confirm that incorporating GAN-generated samples significantly improved CNN performance in deeper waters, reducing RMSE and MAE by approximately 35%–45% in depths beyond 15 m. This enhancement stems from the enriched feature space provided by the synthetic samples, which enables the model to more comprehensively learn the spectral–depth mapping relationship under signal-limited conditions. Compared to the latest HHU24SWDSCS model in Jiuzhang Atoll, the GAN-augmented and stratified approach extends the reliable retrieval depth from 15 to 19 m while also improving accuracy within the 15-m range (Table III).

However, the effectiveness of GAN augmentation is inherently constrained by the quality and representativeness of the original samples. If the source data contain biases or noise, the generated samples may perpetuate or even amplify these errors [51]. Challenges such as mode collapse may also limit the diversity of synthetic data. Future research could explore physics-constrained generative frameworks—for instance, by incorporating radiative transfer-based spectral attenuation rules—to improve the physical consistency and realism of generated samples [52]. Such a hybrid approach, blending data-driven learning with physical principles, holds strong potential for enhancing both the accuracy and interpretability of bathymetric inversion systems.

D. Uncertainty Analysis

Uncertainties in bathymetric inversion arise from multiple sources, including data precision, environmental variability, and model generalization limits. Key error origins can be categorized into optical-environmental noise, sample distribution, geomorphological complexity, and spatiotemporal inconsistencies.

Optical attenuation and optical-environmental noise constitute a fundamental source of uncertainty, particularly in deeper waters. Variability in water constituents—such as suspended sediments, chlorophyll-a, and colored dissolved organic matter (CDOM)—can significantly alter reflectance properties, leading to depth retrieval errors even under identical spectral inputs [53], [54]. Additional uncertainties stem from atmospheric correction residuals, sun-glint effects, sea-surface roughness, and variations in solar geometry, all of which modulate the radiative transfer path [55], [56]. While these factors exert limited influence in clear shallow waters, their impact is magnified in deeper zones where bottom signals are weak and susceptible to environmental noise. The depth-dependent limitation is reflected in the increase of RMSE from 1.36 m in the shallow zone (0–12 m) to 2.61 m in the deep zone (15–19 m), even for the best-performing stratified CNN model. This indicates that optical attenuation accounts for a substantial portion of the accuracy degradation with depth.

Sample distribution and representativeness represent another major contributor to uncertainty, particularly beyond 15-m depth, where ICESat-2 reference data become sparse. The effectiveness of GAN-based augmentation in mitigating this limitation is evidenced by the reduction of RMSE from 3.82 to 2.61 m in the 15–19-m interval when comparing the standard CNN with the GAN-augmented and stratified CNN. This improvement suggests that sample deficiency contributes a comparable magnitude of error to optical attenuation in deep-water inversion.

Seabed geomorphological complexity also imposes a certain influence on inversion accuracy. The coral reefs in the study area are characterized by diverse terrain features, ranging from relatively broad, shallow reef flats to extremely steep reef slopes. Notably, the submerged reef zones feature well-developed spur-and-groove structures, which result in significant spatial variability in seabed slope [57]. As detailed in Section IV-D, the inversion error is significantly higher in steep reef margins (>15°)—typical of these spur-and-groove zones—with an RMSE of 2.78 m, compared to 1.49 m in flat terrains (<5°). This discrepancy is attributed to subpixel heterogeneity, where a single 10-m pixel covering a steep slope encompasses a large vertical depth range, resulting in a mixed spectral signal. In addition, geometric registration errors between satellite imagery and multibeam data are amplified in these high-gradient zones.

Temporal mismatches between ICESat-2, Sentinel-2, and multibeam acquisitions introduce additional uncertainty. Tidal fluctuations, wave setup, and seasonal hydrodynamic processes can cause depth variations at a given location, reducing temporal consistency across datasets [58]. Moreover, the spatial concentration of multibeam data in depth over 7 m creates an imbalanced validation set, potentially skewing error metrics and masking model biases in very shallow or complex substrate environments.

Collectively, the uncertainty analysis indicates that optical attenuation and sample bias are the primary error sources in deep reef waters, while geomorphological complexity acts as a localized error driver in steep margins. These findings highlight the necessity of integrated strategies that jointly address physical signal limitations, training data representativeness, and geometric constraints in future SDB workflows.

E. Applicability to Complex Optical Environments

While this study demonstrates the efficacy of the GAN-CNN framework in the optically clear waters of Jiuzhang Atoll, coral reef ecosystems globally exhibit significant optical diversity. Reefs are not limited to oligotrophic clear waters but are also found in turbid, sediment-rich environments (referenced as Case-2 waters), such as those influenced by river plumes or coastal resuspension [59], [60], [61]. Extending our methodology to these complex environments requires inclusion of additional data constraints of localized environmental features.

First, turbidity significantly alters the radiative transfer process. In Case-2 waters, the presence of suspended par-

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ticulate matter and CDOM increases the diffuse attenuation coefficient, causing a rapid exponential decay of the optical signal [62]. This physical limitation drastically reduces the effective penetration depth of passive multispectral imagery (e.g., Sentinel-2), compressing the radiometric range available for bathymetric inversion [63].

Second, the availability of active training data is similarly constrained by water clarity. Although ICESat-2 possesses superior penetration capabilities compared to passive sensors, its performance is not immune to turbidity. Previous studies have shown that the density of valid bottom-reflected photons decreases significantly as turbidity increases, with signal extinction occurring at much shallower depths compared to clear waters [64], [65]. Consequently, in turbid regions, the distribution of high-quality bathymetric control points becomes skewed toward very shallow depths, leaving deeper areas with sparse or nonexistent ground truth.

Beyond signal attenuation, complex optical environments introduce additional spectral ambiguities. In particular, distinguishing between dark benthic substrates in shallow water and brighter substrates at greater depths remains challenging, as both scenarios can produce similar water-leaving radiance. While the proposed GAN-CNN framework leverages spatial texture information and ICESat-2 supervision to mitigate such ambiguities, the current model primarily infers depth from spectral radiance and spatial patterns and does not explicitly incorporate benthic substrate properties. Consequently, in optically complex environments, GAN-generated samples may still inherit biases associated with clear-water spectral characteristics.

In light of these constraints, the fixed depth stratification thresholds used in this study (e.g., 0–12 m) should not be applied rigidly to all environments. Instead, when extending the proposed framework to Case-2 waters, the stratification scheme must be adaptively redefined based on two local factors: 1) the site-specific water transparency (optical-depth limits) and 2) the statistical distribution of available ICESat-2 training samples. Specifically, the segmentation thresholds should shift to shallower ranges in turbid waters to match the compressed effective depth of both the passive spectral signal and the active LiDAR data. Furthermore, future extensions of this framework could benefit from incorporating benthic habitat or substrate classification as an additional input. Integrating coarse-resolution substrate maps (e.g., sand, coral, and seagrass) would provide explicit albedo priors, enabling a more semantic-aware bathymetric inversion that decouples substrate reflectance from water depth and enhances robustness in optically complex reef systems.

VI. Conclusion

This study establishes an integrated shallow-water bathymetry inversion framework that synergizes multisource remote sensing data—including Sentinel-2 multispectral imagery and ICESat-2 ATL03 photon data—with deep learning techniques. The influence of different data processing and modeling strategies on inversion accuracy is systematically evaluated. The main findings are summarized as follows.

1) Superior Performance of CNN-Based Inversion: CNNs effectively capture the complex nonlinear relationship between multispectral reflectance and water depth across diverse optical environments. Even without stratification, the CNN model consistently outperforms other approaches, such as RFs and BPNNs, demonstrating a competent feature learning capacity in shallow-water bathymetry estimation.

2) Enhanced Accuracy in Deeper Zones via GAN-Augmented Training: To overcome the limitation of sample scarcity at depth over 15 m, a GAN is employed to synthesize physically plausible training samples. The inclusion of GAN-generated data significantly improves CNN model performance in deeper waters, reducing RMSE from 3.70 to 2.40 m and MAE from 3.00 to 1.63 m. This confirms that GAN-based augmentation effectively mitigates training instability and boosts predictive accuracy in data-scanty regions.

3) Improved Spatial Accuracy and Robustness Through Stratified Modeling: By partitioning the study area into distinct depth intervals based on the optical characteristics derived from ICESat-2 data, stratified CNN models are tailored to distinct depth regimes. RMSEs of 1.36, 1.71, and 2.40 m are calculated for the respective layers, outperforming the RMSE of 3.70 m with the nonstratified model. Validation against independent multibeam data yields an overall RMSE of 2.40 m and an MAE of 1.64 m, which remain reasonable given the concentration of multibeam observations at depth >7 m. The stratified strategy not only improves accuracy at deeper depth but also preserved spatial continuity across the bathymetric map.

In summary, this research validates the effectiveness of integrating ICESat-2 LiDAR, Sentinel-2 imagery, and deep learning for high-resolution bathymetric mapping in coral reef environments. The proposed combination of GAN-driven data augmentation and depth-stratified convolutional modeling enables reliable retrieval of water depths up to but not limited to 19 m, with enhanced accuracy and stability across the entire shallow-water domain. This methodology offers a prospective framework for large-area bathymetric mapping in coral reefs worldwide, supporting coastal ecosystem management and habitat monitoring.

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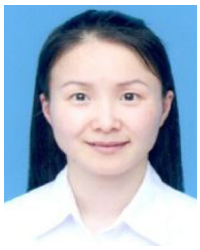
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